

LEDGER · 2026-09-04 · BIOMEDICAL ENGINEERING

Thermoresponsive In-Situ Gelling Biogel via Polyacrylamide/Gelatin Co-Polymerization

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Weaver Ten20 Conductive Paste
Entry application	long-term continuous ambulatory EEG monitoring
Measured advantage	2.36x SNR improvement (25.2 dB vs 10.64 dB) and >4.8x impedance reduction (310 Ω vs 1.5–5.0 kΩ)
Time to first revenue	16 months
Gross margin at parity	87.2%
Startup CapEx	\$215,000
Payback from first sale	0.3 months
Capital productivity	94.01x

Regulatory risk

Moderate — qualification costs reported (\$450,000 estimated)

Why it failed

- Voided by: Form-factor / handling regression, Absence search missing
- Evidence rests on non-peer-reviewed sources
- Absence asserted, search unreported

Assessment

The concept demonstrates clear signal superiority and strong margins, but fails on form factor parity due to requiring refrigerated storage and cold-water/ice removal compared to room-temperature incumbent workflow. Additionally, it lacks a structured Market Absence Ledger section with numeric search counts.

Also flagged

- Price parity asserted, not sourced
- runsheet_no_quantities

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

Terms

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